

Student: Jonatan Rabjerg
Date: 1/15/24

Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 5

Use the Limit Comparison Test to determine if the following series converges or diverges.

(Hint: Limit Comparison with $\sum_{n=1}^{\infty} \frac{1}{n^2}$)

$$\sum_{n=1}^{\infty} \frac{n-7}{n^3 - 2n^2 + 12}$$

Recall the Limit Comparison Test. Suppose that $a_n > 0$, $b_n > 0$, and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$. If $0 < L < \infty$, then $\sum a_n$ and $\sum b_n$ converge or diverge together. If $L = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.

Let $a_n = \frac{n-7}{n^3 - 2n^2 + 12}$. Choose an expression for b_n that is similar to the n th term a_n and is known to converge or diverge.

Look at the largest degree terms in the numerator and denominator of a_n . Determine b_n as follows.

$$b_n = \frac{n}{n^3} \\ = \frac{1}{n^2}$$

Notice that in a_n , the denominator will always be positive for all $n \geq 1$. However, the numerator $n - 7 > 0$ when $n \geq 8$. Therefore, $a_n > 0$ for all $n \geq 8$. Since both a_n and b_n are positive for all $n \geq 8$, the Limit Comparison Theorem can be applied.

Now determine $\frac{a_n}{b_n}$.

$$\frac{a_n}{b_n} = \frac{\left(\frac{n-7}{n^3 - 2n^2 + 12} \right)}{\left(\frac{1}{n^2} \right)} \\ = \left(\frac{n-7}{n^3 - 2n^2 + 12} \right) \left(\frac{n^2}{1} \right) \\ = \frac{n^3 - 7n^2}{n^3 - 2n^2 + 12}$$

Multiply by the reciprocal.

Distribute in the numerator.

Next, divide the numerator and denominator by n^3 .

$$\frac{a_n}{b_n} = \frac{\frac{(n^3 - 7n^2)}{n^3}}{\frac{(n^3 - 2n^2 + 12)}{n^3}} \\ = \frac{1 - \frac{7}{n}}{1 - \frac{2}{n} + \frac{12}{n^3}}$$

Divide and simplify.

Evaluate $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{1 - \frac{7}{n}}{1 - \frac{2}{n} + \frac{12}{n^3}} \\ &= \frac{1 - 0}{1 - 0 + 0} \\ &= 1\end{aligned}$$

Since $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$ and $0 < 1 < \infty$, then by the Limit Comparison Test, $\sum_{n=1}^{\infty} \frac{n-7}{n^3 - 2n^2 + 12}$ and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge or diverge together.

The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges because it is a p-series with $p = 2 > 1$.

Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, $\sum_{n=1}^{\infty} \frac{n-7}{n^3 - 2n^2 + 12}$ converges by the Limit Comparison Test.